

Develop a Student Performance Prediction Tool using K-Nearest Neighbors (KNN) Algorithm

A university wants to develop a simple AI model that predicts whether a student will **PASS** or **FAIL** an exam based on academic behavior and learning habits.

You are provided with a dummy dataset, *student_knn_data-gemini-3.xlsx* to train a Machine Learning model using the K-Nearest Neighbors (KNN) algorithm.

You are required to:

- (i). Load the created excel file into pandas
- (ii). Check for missing or inconsistent values
- (iii). Encode the Final_Result labels (PASS = 1, FAIL = 0)
- (iv). Normalize feature values using MinMaxScaler or StandardScaler from scikit-learn
- (v). Split the data into training (80%) and testing (20%) sets
- (vi). Train a K-Nearest Neighbors (KNN) classifier using the dummy dataset
- (vii). Evaluate the model using:
 - Accuracy
 - Confusion Matrix
 - Classification Report
- (viii). Test K values from 1 to 5
- (ix). Plot Accuracy verses K and choose the best K using matplotlib.

Develop a Python Program to test the AI Model Application

Write a Python program which prompts the user to input: Hours studied, Attendance percentage, number of assignments completed, and past exam score. The system should then predict if the student will **PASS** or **FAIL**.

- (i). Validate that all inputs are numeric and within expected ranges.
- (ii). Apply the same scaling method used during model training.
- (iii). Use the trained KNN model to predict **PASS** or **FAIL**.
- (iv). Display a clear message such as: **The student is likely to PASS**